scientific reports

OPEN

Unveiling community structure, antimicrobial resistance, and virulence factor of a wastewater sample of dairy farm located in mayurbhanj, odisha, India

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Nutrient-rich dairy wastewater (DWW) is an excellent growing medium for microbes. Their antimicrobial resistance (AMR) genes and pathogenic roles remain in the DWW and even multiply in environmental settings, in contrast to many chemical toxins that break down over time. Necessary steps and standardized techniques for tracking AMR in DWW samples are desperately needed. In this context, a DWW sample was evaluated to assess the necessity of remediation and develop a suitable treatment technique. Physicochemical characterizations of the sample showed an elevated level of pollutants like proteins, fats, and carbohydrates that led to the water pollution and microbial diversity (e.g., 36 phyla, 72 classes, 111 orders, 168 families, 275 genera, and 347 species). The Shannon and Simpson indices showed that the DWW sample had a high level of microbial diversity of a few species. The gene ontology (GO) analysis revealed the functional categories with 2795 genes belonging to 11 virulence categories. Most of the identified AMR genes belonged to beta-lactamase, and the majority of them were linked to *Escherichia coli*, *Mycobacterium tuberculosis*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Enterobacter cloacae*, etc. The major bacterial phyla carrying AMR genes included Firmicutes (36%), Proteobacteria (31%), Actinobacteria (21%), and Bacteroidetes (5%).

Keywords Dairy wastewater, Physico-chemical parameters, Metagenomic sequencing, Antimicrobial resistance, Virulence factor

Dairy industries generate excessive waste and by-products while producing a wide range of products, as listed in Table 1[1,2]. Their production capacity has increased exponentially due to the growing demand for dairy products, which reciprocates in discharging a huge DWW. Approximately 11 L of water is required to process 1 L of milk, resulting in 10 L DWW[3,4]. DWW is rich in nutrients like lactose, proteins, amino acids, fats, minerals, etc., resulting in a high concentration of chemical oxygen demand (COD), biological oxygen demand (BOD), suspended solids (SS), oil & grease, total nitrogen (TN), and total phosphorus (TP)[5]. The composition of DWW varies depending on the environmental conditions and wastewater (WW) treatment processes. The production processes of different dairy products undergo separate technological lines, resulting in the fluctuation of quality and quantity of DWW, which creates difficulties in its treatment[6,7]. DWW is non-toxic but contains fats, rich nutrients, and detergents. Hence, it is characterized by a substantially elevated 30–40 °C temperature and high COD. It is slightly alkaline and contains a significant quantity of SS due to curd content, compared to other dairy wastes[8].

In addition to being highly nourishing to people, DWW is an excellent growing medium for microbes[6]. A mixture of bacteria, fungi, and protozoa supports the complex systems of organisms found in the DWW. Microorganisms and their AMR remain in the DWW and even multiply in environmental settings, in contrast to many chemical toxins that break down over time. They persist in the DWW even after its treatment[9]. In environmental microbiology, virulence factors (VFs) and AMR have become major issues, particularly in effluents from the agriculture and food industries. AMR genes and harmful microbes spread quickly through DWW and enter the ecosystem. Numerous classes of AMR genes, including aminoglycosides, beta-lactams,

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Sl. No.	Major dairy products	Wastes
1	Raw milk	Skimmed milk, spoiled milk, and spilled milk contain high content of nitrogen and phosphorous
2	Sterilized and pasteurized milk	WW generated during production and packaging contains SS and organic matters
3	Cheese	Cheese whey waste and whey permeates
4	Butter; clarified ghee; ice-cream	Clarified butter sediment generated during production of butter or clarified ghee
5	Curd	Curd chunks waste
6	Yogurt	Fatty effluents like grease & oil
7	Milk powder	Sludge and WW produced during powder separation

Table 1. List of wastes generated during processing of major dairy products.

aminocoumarins, fluoroquinolones, macrolides, phenicols, tetracyclines, phosphonic acid, disinfecting and antiseptic agents, elfamycin, rifamycin, and multidrug resistance genes, have been identified by whole genome sequencing analysis. AMR poses serious hazards to human health because it can proliferate within host bacteria, spread to other bacterial populations, and evolve further. AMR spreads in the environment through four main mechanisms: selective pressures, genetic mutations, recombination, and horizontal gene transfer. Necessary steps must be taken to lessen the dangers posed by environmental AMR genes. Standardized techniques for tracking AMR in environmental samples are desperately needed to support trend analysis and international comparisons. To lessen AMR-induced environmental contamination, technical solutions should be developed and implemented[10]. DWW is an environmental host for *Escherichia coli*, *Staphylococcus aureus*, *Acinetobacter* spp., and *Pseudomonas* spp. that possess VFs like host adhesion, biofilm formation, motility, invasion, quorum sensing and regulation, secretion systems, siderophores, anti-phagocytosis, and toxin synthesis[11].

Many dairy industries follow on-site WW treatment technology to tackle their DWW; however, its cost burdens revenue generation[12]. Further, the DWW cannot be discarded directly into the public sewers or water resources due to stringent environmental regulations. Therefore, a low-cost WW treatment system must be employed for the DWW treatment to manage it smoothly[1].

Biological treatment is one of the alternative options due to its environmental and cost benefits[13]. Since DWW is rich in protein and fats, indigenous microorganisms decompose them to lessen their pollutant load[6]. Various organic and inorganic nutrients in the DWW create microenvironments that favour the colonization and growth of different microorganisms[14]. These microorganisms directly consume the nutrients of the DWW and produce different metabolites, which react with the nutrients to produce non-toxic compounds. Hence, microorganisms directly consume and indirectly decompose the various nutrients in the DWW, decreasing its high pollutant load. Evaluating microbial diversity and further screening effective microorganisms for the biodegradation of nutrient-rich DWW can develop an efficient bioremediation technology[14,15]. In this article, we have assessed a DWW sample collected from a local dairy industry for the development of its biological treatment process. For this purpose, a detailed physicochemical characterization was done, and it was correlated with its microbial diversity to select some efficient microorganisms. Its microbial diversity was analysed using whole-genome and metagenomic sequencing techniques.

Materials and methods

Sample collection and characterizations

The real DWW sample for the experiment was collected from the first WW collecting pond (before the WW treatment plant (Fig. 1(b)) of a dairy industry located in the Mayurbhanj district of Odisha, India (Fig. 1(a)), by following all environmental and biological precautions required to collect the WW for the research purpose[15]. Its temperature and pH were measured at the time of collection using a digital thermometer and a portable pH meter, respectively. It was collected in a sterile Borosil glass bottle (1 L), as shown in Fig. 1(c). The WW sample bottle was packed in ice (4 °C) during transportation to the laboratory and subsequently stored at -20 °C for further analysis. Physicochemical characterizations like electrical conductivity (EC), COD, BOD, total suspended solids (TSS), silicate, oil & grease, sulfate, TP, and TN of the DWW sample were conducted by following the standard methods[16].

Deoxyribonucleic acid (DNA) extraction

The whole genome and metagenomic sequencing study analysed the biological properties of the DWW sample. For this purpose, the Qiage-DNeasy-Power-soil-Pro-kit was used to extract the genomic DNA from the DWW by following its manufacturing instructions. Instead of the PowerWater DNA Isolation Kit, we used the PowerSoil DNA Isolation Kit because the PowerSoil DNA Isolation Kit is so effective at handling materials with high particulate matter, which are possible polymerase chain reaction (PCR) inhibitors[17]. The DWW sample used in this study was not clear water; instead, it was turbid and full of organic debris, SS, and microbial biomass, which are traits more common to sludge or soil samples than to clear water samples. Due to filter blockage and ineffective analysis in the presence of particulate matter, a previous study revealed that the PowerWater Kit produced variable findings and lower DNA concentrations[17]. The PowerSoil Kit eliminates inhibitors like humic substances that are prevalent in complicated environmental samples, resulting in effective analysis of the microbial cells. The Qiagen Powersoil prokit has been used for water samples in several studies[50,51], rather than the PowerWater kit, as the lysis buffer is capable of lysing several microbial types, ensuring maximum representation and is ideal for metagenome studies. Walden et al. used different kits to extract DNA from water and wastewater

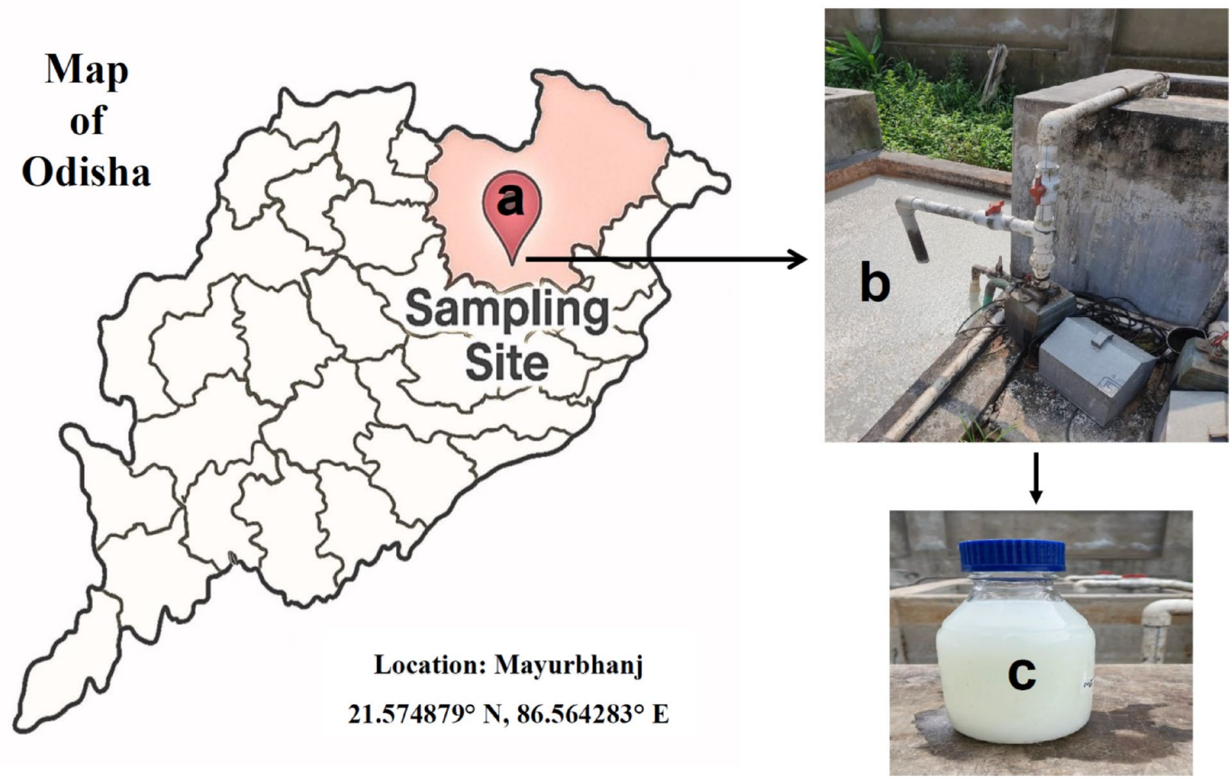

Fig. 1. Exact location of the sampling site (**a**); DWW in first collecting pond (**b**); DWW sample collected in a sterile Borosil glass bottle (**c**).

Barcode sequence	AAGGATTCATTCCCACGGTAACAC
Qubit	11.3
Yield	226

Table 2. Barcode details of the genomic DNA.

(WW) samples[17]. The Power water kit showed the least efficiency, while the Power-soil kit (predecessor of the Power-soil-Pro-kit) showed the most efficiency. Yet in another study[52]the Powersoil pro kit was superior to the Power water kit in providing the highest representation of organisms in marine water samples. This study tried to detect *Haliotis cracherodii*, a critically endangered marine mollusc species. The amplification of *H. cracherodii* was detected only in the PowerSoil pro kit extracted samples, while the Power Water kit could not do so. Consequently, the PowerSoil Kit was more suitable for our particular sample type to guarantee consistent, high-quality DNA extraction adequate for downstream metagenomic analysis. A spectrophotometer was used to measure the concentration and purity of extracted DNA (Nanodrop-2000, ThermoFisher Scientific). Its integrity was assessed with the help of agarose gel electrophoresis, and its quantity was measured using the Qubit-dsDNA-HS-assay-kit. It was subsequently preserved at -20 °C for further analysis.

Whole-genome and metagenomic sequencing

Library preparation

The genomic DNA library was prepared using the NEBNext-ultra-end-repair-kit. The end-prepared sample was barcoded using Blunt-TA-ligase-master-mix. The details of the barcode are listed in Table 2. The equimolar concentration of the barcoded sample was pooled, and the library was prepared by ligating sequencing adapters onto double-stranded DNA fragments using NEBQuick-DNA-Ligase as per the manufacturer's protocol. After purification with AMPure-XPbeads, the prepared library was ready for library quality control.

Raw data processing

DNA isolated from the DWW sample was subjected to whole metagenome shotgun sequencing to characterize the microbial population and its functional potential. The Oxford-Nanopore-Technologies-PromethION platform, which offers long-read sequencing appropriate for investigating diverse microbial communities in environmental samples, performed the shotgun metagenomic sequencing on the whole genomic DNA. The raw reads were checked qualitatively using NanoStat (https://pypi.org/project/NanoStat/) for further assessment.

The removal of adapter sequences during pre-processing was performed using the Porechop tool (https://github.com/rrwick/Porechop). Adapters at the ends of reads were trimmed off. Adapters in the middle of a read were considered chimeric and separated into different distinct reads. After getting several high-quality reads, they were compared against the non-redundant protein database using the distributed alignment matrix for oligonucleotide databases (DIAMOND) basic local alignment search tool (BLAST)[18].

Taxonomic classification
Functional classification was performed by aligning the sequences against the non-redundant protein database of the National Center for Biotechnology Information (NCBI) using the DIAMOND BLAST, with the default E-value cutoff of ≤ 0.001. Then, the metagenome analyser for long reads (MEGAN-LR) graphical user interface was used to estimate and interactively explore the taxonomical content[19]. The comparison of reads was done against the taxonomic information. The resulting BLAST files were converted into MEGAN-compatible format using the long-read specific algorithm, enabling functional annotation, including GO terms, software for environment and ecosystem database (SEED) analysis, and evolutionary genealogy of genes: non-supervised orthologous groups (EGGNOG) classifications[20]. This approach allows for assigning functional categories even for reads supported by a single count, maximizing the depth of functional annotation across multiple databases. The MEGAN interactive mode analysis was performed on the compared reads, and then the read abundance values were calculated from phylum to species. For taxonomy assignment in MEGAN, several key parameters are typically used, such as E-value cutoff, **the** lowest common ancestor (LCA) algorithm to map the precise taxonomy, and relative abundance calculation within the sample to assign and visualize taxonomy at all levels, from domain to species, based on DIAMOND-generated BLAST results. For the classified species and their tax IDs (based on NCBI taxonomy), the taxonomy lineage of the species was mapped using the Krona tool. The Krona tool was used to generate the Krona chart to generate sample-wise Krona visualizations that allowed the examination of relative abundance and confidence in the complicated hierarchies of metagenomic divisions. The interactive nature of these plots gave the quantitative phylogenetic information.

GO classification
GO defines the universe of concepts relating to gene functions and their relations with each other. The GO classification was performed using MEGAN-LR[19]. Genes were mapped onto the InterPro identifiers and placed in a GO-based classification. The functional results were categorized based on the MEGAN results. Further, based on the BLAST results against the non-redundant protein database, the EGGNOG database was used for the DWW sample[21]. All core orthologue sequences were carefully grouped into clusters of orthologous groups (COGs) of proteins using EGGNOG. In addition, SEED analysis was performed as it categorized the genes into specific functional roles, compared the metagenomic sequences to the database, and assigned functional roles to the genes or gene fragments.

Diversity analysis
The microbiological diversity measured in a sample, which represents its taxonomic diversity, is known as alpha diversity. Diversity indices were calculated using the *vegan* R package with a species-level abundance matrix. Shannon (using a natural logarithm base) and Simpson indices were computed row-wise for alpha diversity for each sample. The richness and evenness of a microbial community within a sample are measured statistically by the alpha diversity indices. Beta diversity, representing the differences in species composition between samples, was analysed using the *betadiver* function from the *vegan* package (https://CRAN.R-project.org/package=vegan). Specifically, the 'z' index was used to calculate the dissimilarities based on the proportion of shared and unique species between communities. Beta diversity was assessed without reordering or squaring the distance measures.

AMR analysis
Metagenomic sequencing of the whole genome was conducted to identify AMR and VFs. This enables the high-throughput investigation of complete microbial communities in DWW, enabling the discovery of known and novel virulence genes (VGs) without prior knowledge of the species present. The entire sample was directly sequenced and represented in the whole-genome metagenomic sequencing. The AMR analysis used the comprehensive antibiotic resistance database (CARD) (https://card.mcmaster.ca/). The AMR gene sequences were retrieved from the CARD and indexed using the NCBI BLAST (version: NCBI-blast-2.2.29+). The processed reads from each sample were then compared against the NCBI BLAST-indexed AMR sequences. Using a sequence homology approach, hits with a minimum of 50% identity to the CARD reference sequences were considered the potential AMR gene matches. The associated protein accession, DNA accession, drug class, AMR gene name, AMR gene family, and resistance mechanism were reported for each input read that matched an AMR reference sequence. This study provided comprehensive information on the specific AMR gene and its resistance mechanism.

VF analysis
The virulence factor database (VFDB) was used for the virulence analysis in this study. The VGs sequences were retrieved from the VFDB and indexed using the NCBI BLAST (version: NCBI-blast-2.2.29+). The processed reads from each sample were compared against the NCBI BLAST-indexed VFDB sequences. Using a sequence homology approach, hits with a minimum of 50% identity to the reference sequences were considered potential VG matches. Input reads that matched reference sequences, like associated VF, pathogenic bacteria, and specific VGs, were reported. This provided comprehensive information on the particular VGs or species.

Each experiment was reproduced with a standard deviation of error within $\pm 3\%$. The strict data processing and analysis steps were performed. All analysis runs used the same versions of all tools (e.g., DIAMOND, MEGAN, etc.) with consistent parameters (e.g., E-value cut-offs, alignment settings, etc.). This reduced the chance of variability arising from software discrepancies.

Results and discussion

Physico-chemical properties

Upon physical observation, the real DWW sample was milky white (Fig. 1c) and had an unpleasant odour and a turbid texture. This may be due to mixing milky whey and casein proteins during the processing/cleaning[22]. The result of the physicochemical characterizations of the DWW sample is given in Table 3. As mentioned, the pH of the DWW sample was 7.88 ± 0.39, which showed that it was slightly alkaline. Since natural products are rich in amino acids and proteins, nitrogen increases the pH slightly above that of pure water[23]. The temperature of the DWW was measured as 27.0 ± 2.0 °C due to several factors like local ambient temperature, seasonal variation, physicochemical properties of WW, and biological fluctuations[24]. The EC (320 ± 14.6 µS/cm) was much lower than the permissible limit. BOD and COD are two important WW quality parameters as they show the level of organic pollution load in the waste effluent. The DWW sample BOD and COD values were 312 ± 13.8 and 750 ± 34.5 mg/L, respectively, which were much higher than the permissible value of [25]. Since milk contains significant organic matters like proteins, fats, and carbohydrates, they contribute to the elevated BOD and COD in the DWW. Further, the microorganisms growing in the waste effluent also contributed to the elevated BOD level. The biodegradability of WW is decided by the BOD/COD ratio[3,53]. If the BOD/COD ratio > 0.5, the WW sample can be successfully handled biologically and is reasonably biodegradable. However, if a WW sample's BOD/COD ratio is between 0.3 and 0.5, microbial seeding is required for biodegradation. If the BOD/COD ratio < 0.3, biodegradation of the WW sample is not advised because exclusive chemically degradable organic substances inhibit the metabolic activity of the biodegrading microorganisms and show refractory properties[26,53–58]. However, there are critics of the correlation between the BOD/COD ratio and biodegradability[27]. The BOD/COD ratio of the DWW sample of the present study was calculated to be 0.41, revealing the presence of higher pollutant levels and poor biodegradability. This DWW cannot be discharged directly into the public sewer system before WW treatment.

In DWW, nitrogen is enriched from amino groups of proteins, and phosphorus is mostly inorganic phosphate, diphosphate, and a minimal amount of organic phosphate. Concentrations of TP and TN are crucial metrics for evaluating the quality of WW streams and serve as the standards for controlling the water quality in various WW treatment plants. The coagulated milk, flavoring additives, and cheese curd particles are the sources of SS in DWW[28]. TSS is a significant parameter used to evaluate WW's strength and determine the efficiency of WW treatment units. It reduces the light penetration, increases the turbidity, and clogs fish gills[6]. This study found that TSS, TN, and TP were 190 ± 8.62, 45.47 ± 2.35, and 10.11 ± 0.61 mg/L, respectively. The observed values of TP and TN demonstrated the risk of eutrophication if they were released into the water resources without proper treatment. The oil & grease content of the DWW was found to be 8.7 ± 0.45 mg/L. Silicate (1.3 ± 0.06 mg/L) and sulfate (14.94 ± 0.76 mg/L) content of the DWW were much below the standard limit. After evaluating the values of the above WW quality parameters, this DWW can create problems like the emission of volatile toxic gases, rapid depletion of dissolved oxygen, a threat to aquatic lives, anaerobic growth of pathogens, eutrophication, and other environmental risk consequences upon discharging it directly into the public water resources[28].

DNA quantification

Results of the concentration, purity, and integrity of the genomic DNA are given in Table 4. Using Nanodrop and Qubit methods, the measured DNA concentration was above 100 ng/µl. This result showed that this genomic DNA sample could be further evaluated by molecular biology techniques like PCR, metagenomic sequencing, and cloning[29].

Data statistics

After preparing the library, the data were further processed to remove adapters using long-read sequencing technology. The Oxford Nanopore Technologies PromethION platform was used for the sequencing study. Using

Parameters	Effluents Discharge Indian Standards (The Environment Protection Rules, 1986)	Characteristics
Colour	-	Milky white
Temperature, °C	45	27.0 ± 2.0
pH	5.5-9.0	7.88 ± 0.39
EC (µS/cm)	-	320 ± 14.6
COD (mg/L)	250	750 ± 34.5
BOD (mg/L)	30	312 ± 13.8
TSS (mg/L)	100	190 ± 8.62
Silicate (mg/L)	-	1.3 ± 0.06
Sulfate (mg/L)	1000	14.94 ± 0.76
TN (mg/L)	110	45.47 ± 2.35
TP (mg/L)	5	10.11 ± 0.61
Oil & Grease (mg/L)	10	8.7 ± 0.45

Table 3. Physico-chemical properties of the DWW sample.

Estimated DNA concentration and purity		
Nanodrop QC	DNA concentration	103.7 ng/µl
	Yield	2592.5 ng
Qubit QC	DNA concentration	104.8 ng/µl
	yield	2620 ng
	QC yield	Optimal
	QC integrity	Intact band

Table 4. Purity and concentration of the extracted genomic DNA.

Sample type	Dairy waste water
Sequencer used	Nanopore PromethION system
Barcode sequence used	AAGGATTCATTCCCACGGTAACAC
Raw reads	2,211,670
Processed reads	2,194,213
Length- within 800–1200 bp	605,556

Table 5. Details of the processing of Raw reads.

Species	Read counts
Mitsuokella multacida	2199
Coriobacteriaceae bacterium	1682
Astraeus odoratus	745
Klebsiella pneumoniae	740
Lactococcus lactis	521
Firmicutes bacterium	511
Verrucomicrobia bacterium	492
Paenirhodobacter enshiensis	430
Rubrivivax gelatinosus	410
Lupinus albus	395
Rhodocyclus tenuis	378
Acidobacteria bacterium	372
Chlorella sorokiniana	362
Selenomonas ruminantium	286
Veillonella magna	264
Micromonospora	237
Prevotella paludivivens	231
Mitsuokella jalaludinii	221
Megasphaera paucivorans	209
Aminicella lysimetric	196

Table 6. Taxonomy classification at the species level (top-20).

the Nanopore PromethION platform, whole metagenome sequencing produced 2,194,213 processed reads, with a median read length that typically falls within a range of around 800–1200 bp and a total data volume of about 4.39 Gbps for the sequenced samples. Sequencing depth varied depending on the diversity of the species being sequenced. The average coverage depth attained by this sequencing was 1.5x. The details of the result during the processing of raw reads are given in Table 5. The processed reads were high-quality data that were used for further microbiome identifications.

Taxonomic profile of microbial communities

A taxonomy analysis of the high-quality reads was performed using MEGAN-LR. The assignment of individual reads to taxa based on their hits was analysed and then summarized from kingdom to species level, along with their read counts. The names of twenty species based on their read counts are given in Table 6. The DWW harboured a complex microbial community with microorganisms belonging to 36 phyla, 72 classes, 111 orders, 168 families, 275 genera, and 347 species. The metagenomic study of the DWW highlighted the intricate interactions among different microbial communities and environmental elements, which indicated the ecological health of the water body in and around the location of the dairy industry. Microbial communities and organic pollutants are deeply interrelated. Many of the heterotrophic microorganisms need organic carbon sources for their metabolic activities. Since the DWW sample was characterised by a high level of BOD and COD, it provided more nutrients for the microbial colonisation and growth, resulting in microbial communities with 347 species. Figure 2 shows the kingdom level of the absolute abundance of microorganisms. Bacterial communities (94.5%) were found to be predominant over other microbial communities {Archaea (1.7%), Eukaryota (3.2%), and Viruses (0.3%)} in the DWW sample. The results showed that most of the reads were classified into Firmicutes, Actinobacteria, and Bacteroidetes phyla, which are most abundant in the DWW sample. Alone, the anaerobe bacterium *Mitsuokella multacida* was assigned a maximum number of reads (around 50%).

Taxonomic profile of bacterial community

At the phylum level, Firmicutes were the most dominant bacteria (51%), as shown in Fig. 3. This was attributed to the rich organic matter of the DWW[20]. Other abundant phyla were Proteobacteria (16%), Actinobacteria (15%), Bacteroidetes (8%), Synergistetes (1.2%), and Verrucomicrobia (1%). At the class level, Negativicutes (25%), Clostridia (11%), Bacilli (10%), Coriobacteriia (8.5%), Bacteroidia (8%), Actinobacteria (8%), Betaproteobacteria (6%), Gammaproteobacteria (5%), Alphaproteobacteria (3%), and Deltaproteobacteria (1%) were found. The bacterial family, such as Selenomonadaceae (22%), Streptococcaceae (10%), Veillonellaceae (6%), Prevotellaceae (5%), Enterobacteriaceae (4%), Bifidobacteriaceae (4%), Atopobiaceae (3%), Coriobacteriaceae (2%), Lachnospiraceae (2%), and Microbacteriaceae (2%), were abundant in the DWW. Numerous researchers have become interested in Firmicutes due to their diverse ecological range and unique physiological, biochemical, and molecular characteristics. Firmicutes degrade organic matter to produce different fatty acids as fermented products during the metabolic period. However, many genera of the Firmicutes produce different antibiotics. Also, some are pathogenic to humans and animals, causing diseases like botulism, tetanus, and anthrax[30]. Because of their metabolic adaptability, Proteobacteria can flourish in various environmental settings. Different anthropogenic activities may be attributed to the dominance of Proteobacteria, Actinobacteria, and Bacteroidetes[31]. Proteobacteria are the most abundant phylum in biofilm formation. In this study, Betaproteobacteria and Gammaproteobacteria were the abundant families of Proteobacteria; they are capable of ammonia oxidation and denitrification. The Comamonadaceae family is a denitrifying bacterium. Most of the anaerobes and micro-anaerobes in the Flavobacteriaceae family are known to break down organic matter and participate in the biogeochemical cycle[14]. A lactate-utilizing anaerobic bacterium, Mitsuokella multiracial, is known primarily for its phytase activity[32]. Hydrogenophaga is related to hydrogen autotrophic denitrification[33].

Taxonomic profile of archaeal community

Phylum Euryarchaeota (2%) was the only archaea among the microbial communities. Methanosarcinaceae, Methanobacteriaceae, Methanoregulaceae, and Methanosaetaceae are dominant families. Archaea like *Methanoregula boonei*, *Methanospirillum*, *Methanosaeta*, and *Methanomassiliicoccales* were found in the DWW sample.

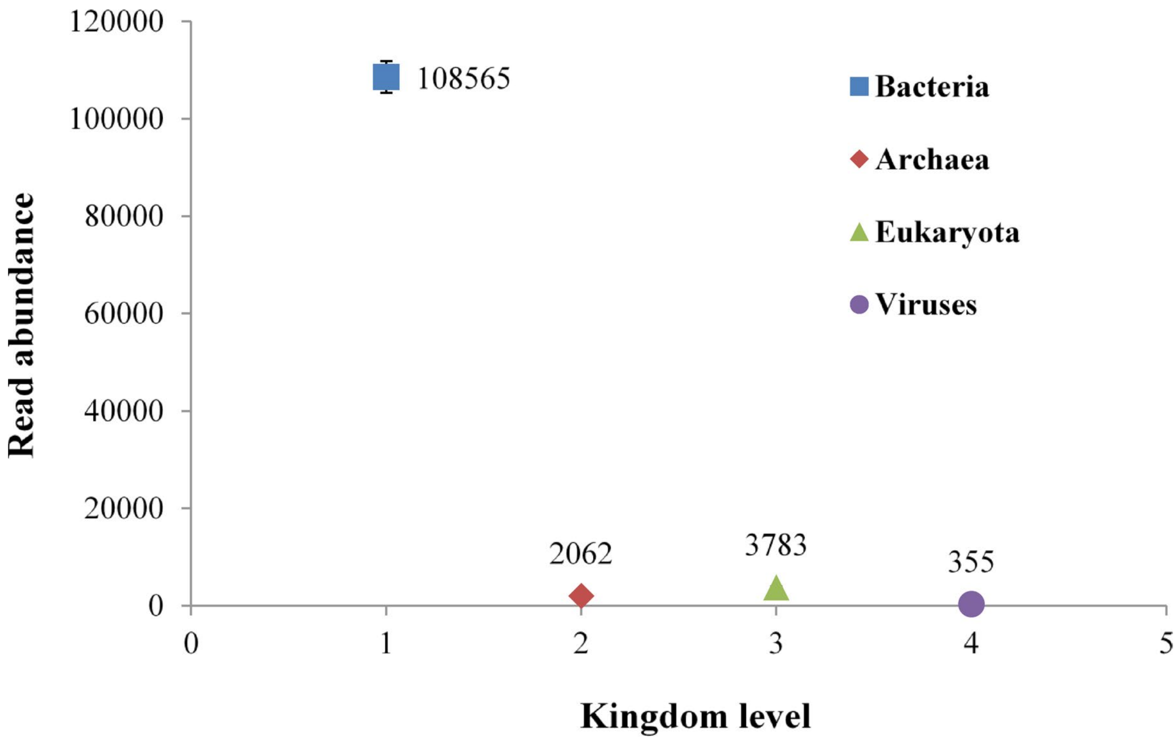

Fig. 2. MEGAN-LR GUI-based taxonomy analysis of the kingdom level absolute abundance of DWW sample.

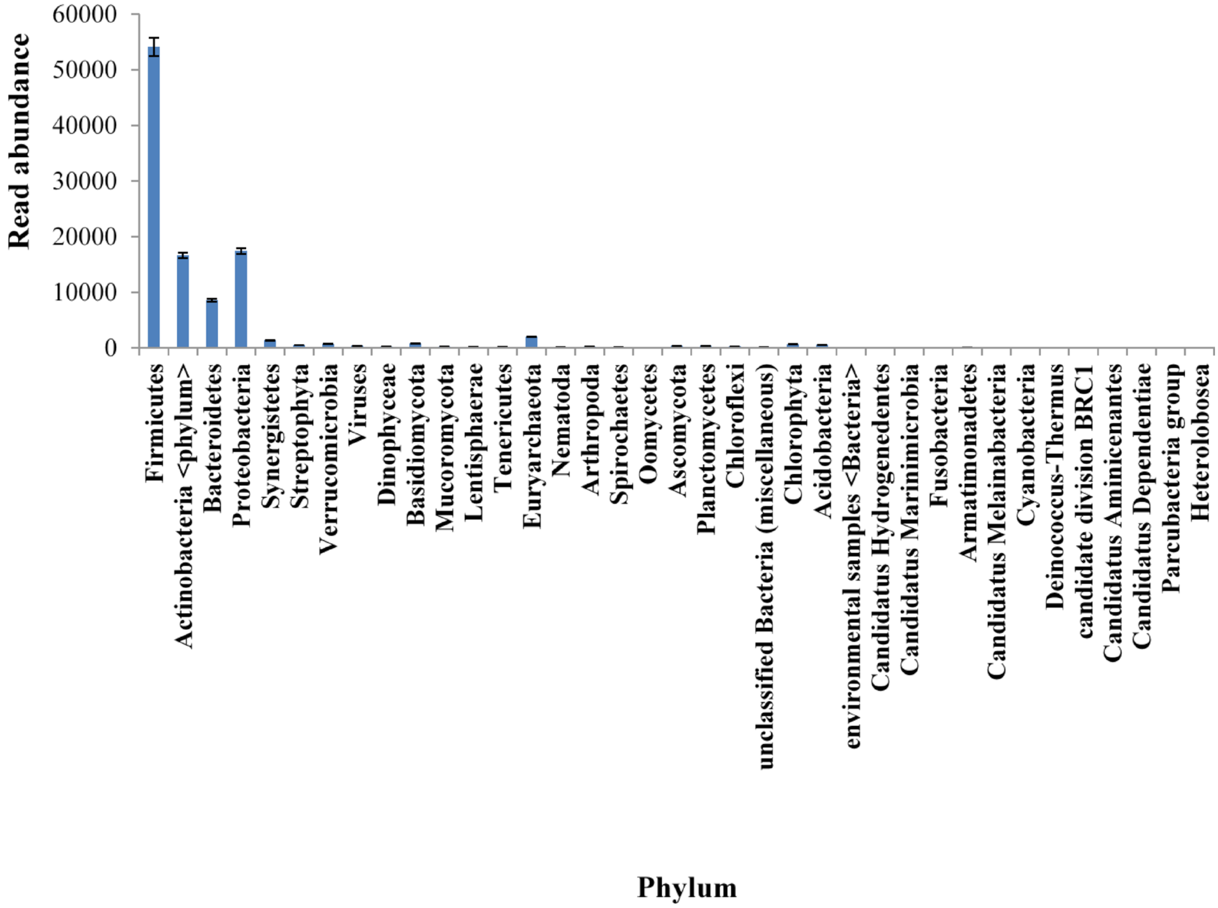

Fig. 3. Phylum level microbial community composition of DWW sample.

Taxonomic profile of eukaryota community
Fungal diversity of Basidiomycota, Ascomycota, and Mucoromycota phylum were found in the DWW sample. Other Eukaryota families like Streptophyta, Chlorophyta, and Mucoromycota were also found. Fungi species like *Astraeus odoratus*, *Beauveria bassiana*, *Aureobasidium melanogenum*, *Rhizoctonia solani*, *Astraeus odoratus*, *Mortierella gamsii*, *Mortierella sp. GBA39*, *Mucoraceae*, *Rhizopus oryzae*, and *Galactomyces candidum* were found in the DWW sample. Some other Eukaryota species in the sample were *Naegleria*, *Trichuris trichiura*, *Symbiodinium microadriaticum*, and *Lupinus albus*. Algal species like *Chlorella sorokiniana*, *Auxenochlorella protothecoides*, *cyanobacteria*, *Helicosporidium sp.* ATCC 50,920, and *Scenedesmus* were found in the DWW.

Taxonomic profile of viral community
Unclassified viral species like *Siphoviridae*, *Myoviridae*, *Podoviridae*, *Caudovirales*, and some unidentified phages were identified in the DWW sample.

Relative abundance of microbial community
Firmicutes were abundantly present in the DWW sample with a relative abundance of 50%, followed by Proteobacteria (16%), Actinobacteria (15%), Bacteroidetes (8%), Euryarchaeota (1%), and Synergistetes (1%). The relative abundance of the microbial community at the species level were as *Mitsuokella multacida* (9%), *Coriobacteriaceae bacterium* (7%), *Astraeus odoratus* (3%), *Klebsiella pneumoniae* (3%), *Lactococcus lactis* (2%), *Firmicutes bacterium* (2%), *Verrucomicrobia bacterium* (2%), *Paenirhodobacter enshiensis* (1.8%), *Rubrivivax gelatinosus* (1.7%), *Lupinus albus* (1.6%), *Rhodocyclus tenuis* (1.6%), *Acidobacteria bacterium* (1.5%), *Chlorella sorokiniana* (1.5%), *Selenomonas ruminantium* (1.2%), and *Veillonella magna* (1%). Numerous distinct microbial communities were also detected in the DWW sample with a relative abundance of less than 1% (Fig. 4). Metagenomic sequencing results were in agreement with other scientific literature data showing the abundance of Firmicutes, Bacteroidetes, and Actinobacteria, representing the major microbial communities at the phylum level[34].

Microbial diversity
The alpha diversity of the microbes characterises the taxonomic diversity. Alpha diversity metrics are measured based on the richness or evenness of a community or a combination of both. A microbial community structure can be summarized using the alpha diversity. Shannon and Simpson indices are the estimators of taxa, combining

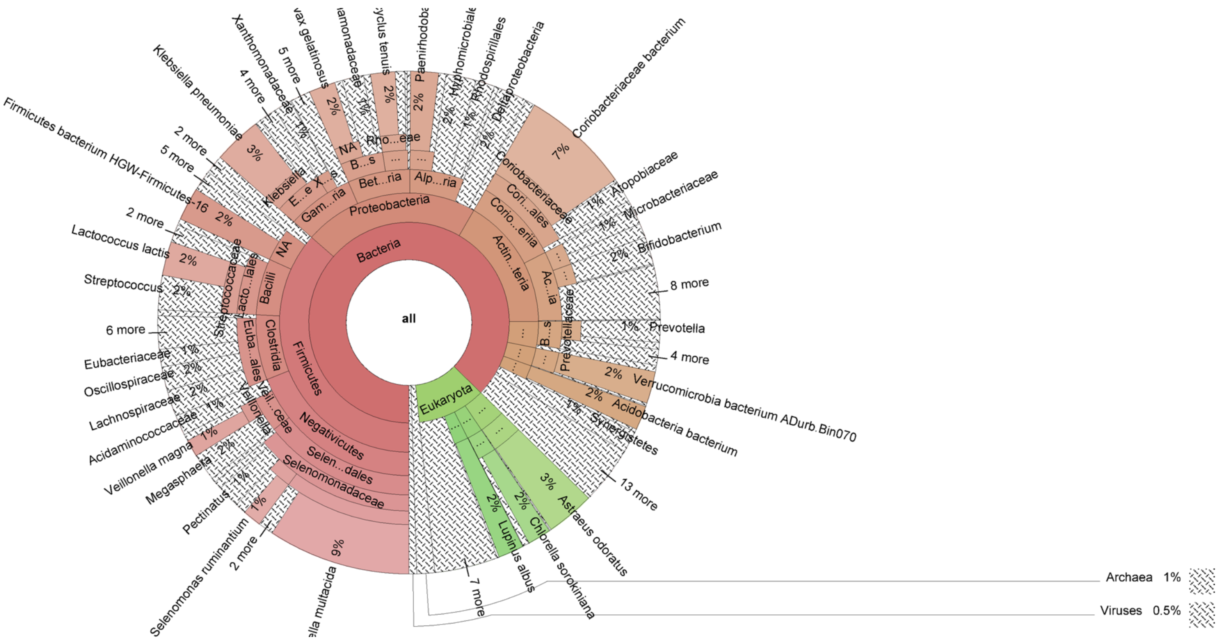

Fig. 4. Mapped relative abundance of microbial community of the DWW sample by using Krona plot.

richness and evenness. Since the Shannon index uses the relative abundances of various taxa, diversity is influenced by taxonomic richness and evenness of microorganism distribution across taxa. It influences the richness of taxa. The Simpson index shows taxonomic dominance and the likelihood that two members of the same taxonomic group will be selected randomly. It ranges from 0 to 1, and as diversity declines, the index rises[35]. The alpha diversity of the DWW sample revealed that the microbial communities exhibited a high diversity of different taxa. The Shannon index was estimated to be 4.89, which showed that the DWW sample has a high level of microbial diversity. Simpson index was estimated to be 0.98, which indicated that the DWW sample contains a very high level of microbial diversity of a few species, which justified the relative abundance of data.

GO classification

Functional GO classification shows its categories and the reads corresponding to each category. The top-20 levels 2 & 3 GO classifications are shown in Figs. 5 and 6, respectively. In the GO category level-1, a total of 78,043 hits were obtained according to the three categories: biological process (32128), molecular function (31955), and cellular component (13960). In the GO category level-2, a total of 481 hits were found for molecular function, biological process, and cellular component. The proportion of genes, such as biosynthetic process (9%), transferase activity (8%), transport (8%), hydrolase activity (7%), nucleotide binding (7%), transporter activity (7%), oxidoreductase activity (6%), membrane (5%), oxidation-reduction process (5%), and nucleic acid binding (5%), contributed to the top-10 category level-2. In the GO category level-3, a total of 3151 hits were found for molecular function, cellular component, and biological process categories. The proportion of genes, such as ion transport (14%), hydrolase activity (9%), ATPase activity (9%), aminoacyl-tRNA ligase activity (7%), cytochrome c oxidase subunit I (6%), peptidase activity (6%), Major facilitator superfamily (6%), kinase activity (5%), transferring acyl groups (5%), and acting on glycosyl bonds (5%), contributed to the top-10 category level-3. GO classification results provided a comprehensive picture of the microbial communities' ecological roles and metabolic capacities in the DWW sample. The different paths that have been discovered may reflect human influences on microbial communities' composition and functionality[31].

COG analysis

EGGNOG provides insights into the functional roles of genes and their evolutionary relationships. In the DWW sample, maximum predominant unigenes were found across various COG categories, such as [E]-transportation of amino acids and their metabolism, [L]-replication, repair, and recombination, [G]-transportation of carbohydrates and their metabolism, [C]-conversion and production of energy, [M]-biogenesis of cell wall, [P]-transportation of inorganic ions and their metabolism, [J]-ribosomal structure, translation, and biogenesis, etc. The distribution of different COG categories, as shown in Fig. 7, defines the microbial diversity, indicating the complex interactions among different microbial populations and their habitats. The COG categories related to [E], [G], and [C] described the fundamental roles of microorganisms in sustaining the biogeochemical metabolism processes and nutrient digestion mechanism[36]. Genes involved in the cellular processes were found to be [L], [M], and [D], showing the existence of active biological mechanisms in play, possibly in reaction to mutations or surrounding environmental pressures. This result further revealed that the microbes constantly adapted, changing their cellular processes to maximize survival and efficacy in various conditions[37].

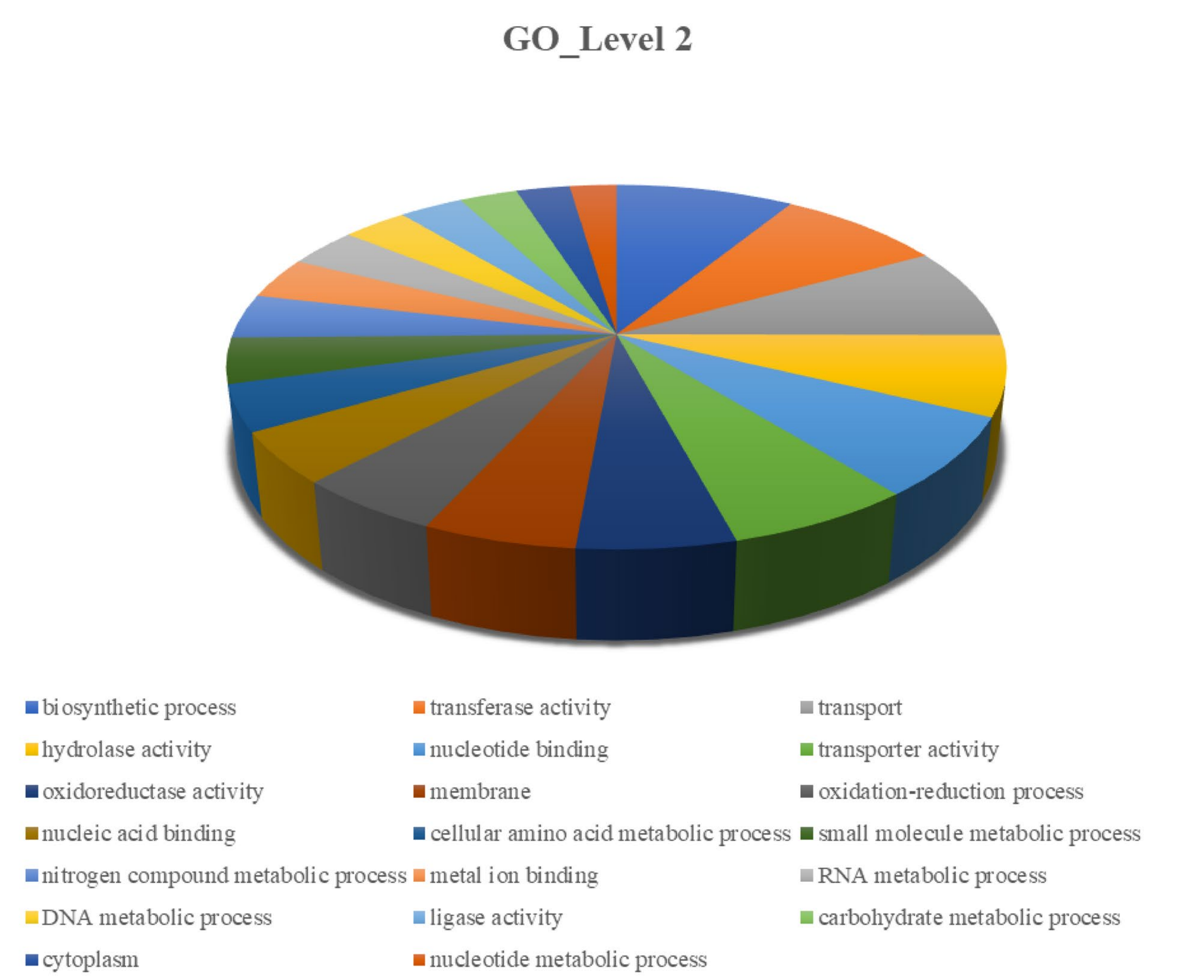

Fig. 5. Top-20 level-2 microbial functional gene ontology classification.

SEED classification

SEED classification helps MEGAN enable the functional behaviour of metagenomic data sets obtained using the DIAMOND BLAST. It assigns different genes to corresponding functional roles, grouped into subsystems[38]. In level-1 SEED analysis, categories like DNA metabolism; carbohydrates; iron acquisition and metabolism; protein metabolism; metabolism of aromatic compounds; stress response; polyamines; cofactors, pigments, prosthetic groups, and vitamins; nitrogen metabolism; respiration; nucleosides and nucleotides; chemotaxis and motility; isoprenoids, lipids, and fatty acids; phosphorus metabolism; transposable elements, prophages, phages, and plasmids; regulation and cell signalling; virulence; dormancy and sporulation; cell wall and capsule; amino acids and derivatives; membrane transport; sulfur metabolism; virulence, disease and defense; ribosomal ribonucleic acid (RNA) metabolism; central metabolism and photosynthesis categories were identified (Fig. 8). In level-2 SEED, categories like universal GTPases; pyruvate metabolism I; glycolysis and gluconeogenesis; glycogen metabolism; branched-chain amino acid biosynthesis; protein chaperones; bacterial RNA processing and degradation; bacterial DNA repair; high affinity phosphate transporter and control of phosphate regulon; pyridoxin (vitamin B6) biosynthesis; DNA repair and UvrABC system; fermentations: mixed acid; adenosine triphosphate (ATP) dependent Type II DNA topoisomerases; anaplerotic reactions and phosphoenolpyruvate (PEP); pentose phosphate pathway; and ribonucleotide reduction were more prevalence. According to a previous research finding, the most prevalent phyla, Proteobacteria, Firmicutes, Bacteroidetes, and Actinobacteria, may be linked to their participation in several ecological processes, including membrane transport, stress response system, protein metabolism, food cycle, and carbohydrate metabolism[20].

AMR gene abundance

A total of 155 resistance gene families, with 100 to 50% identical matches, were detected from the DWW sample (Table 7). Diverse categories of AMR genes, such as rifamycin, carbapenem, and cephalosporin, were found. This outcome provided a critical insight into environmental persistence and dissemination. The most predominant

Fig. 6. Top-20 level-3 microbial functional gene ontology classification.

AMR gene families were resistance-nodulation-cell division (RND) antibiotic efflux pump, rifamycin-resistant EF-Tu, ATP-binding cassette (ABC) antibiotic efflux pump, fluoroquinolone-resistant gyrA, rifamycin-resistant beta-subunit of RNA polymerase (rpoB), major facilitator superfamily (MFS) antibiotic efflux pump, etc. They were linked to carbapenem, aminoglycoside, fluoroquinolone, tetracycline, rifamycin, and cephalosporin drug resistance. For a better understanding of dairy production unit environment contamination, the most contributing taxon was evaluated for the encoding AMR genes family, which linked to *Staphylococcus aureus*, *Escherichia coli*, *Klebsiella pneumoniae*, *Mycobacterium tuberculosis*, *Enterobacter* sp., and *Pseudomonas aeruginosa*. The details of the AMR genes and linked organisms are given in Table 7. The AMR analysis indicated that a diverse group of microorganisms was present in the DWW, indicating there were complex interactions of parameters like environmental conditions, local antibiotic usage patterns, and genetic exchange mechanisms. An essential understanding of the regional dynamics of antibiotic resistance was made possible by the diversity and distribution of AMR genes throughout the sites under study. The results demonstrated the need for focused monitoring and stewardship initiatives to track and control the expansion of AMR in a particular area[37].

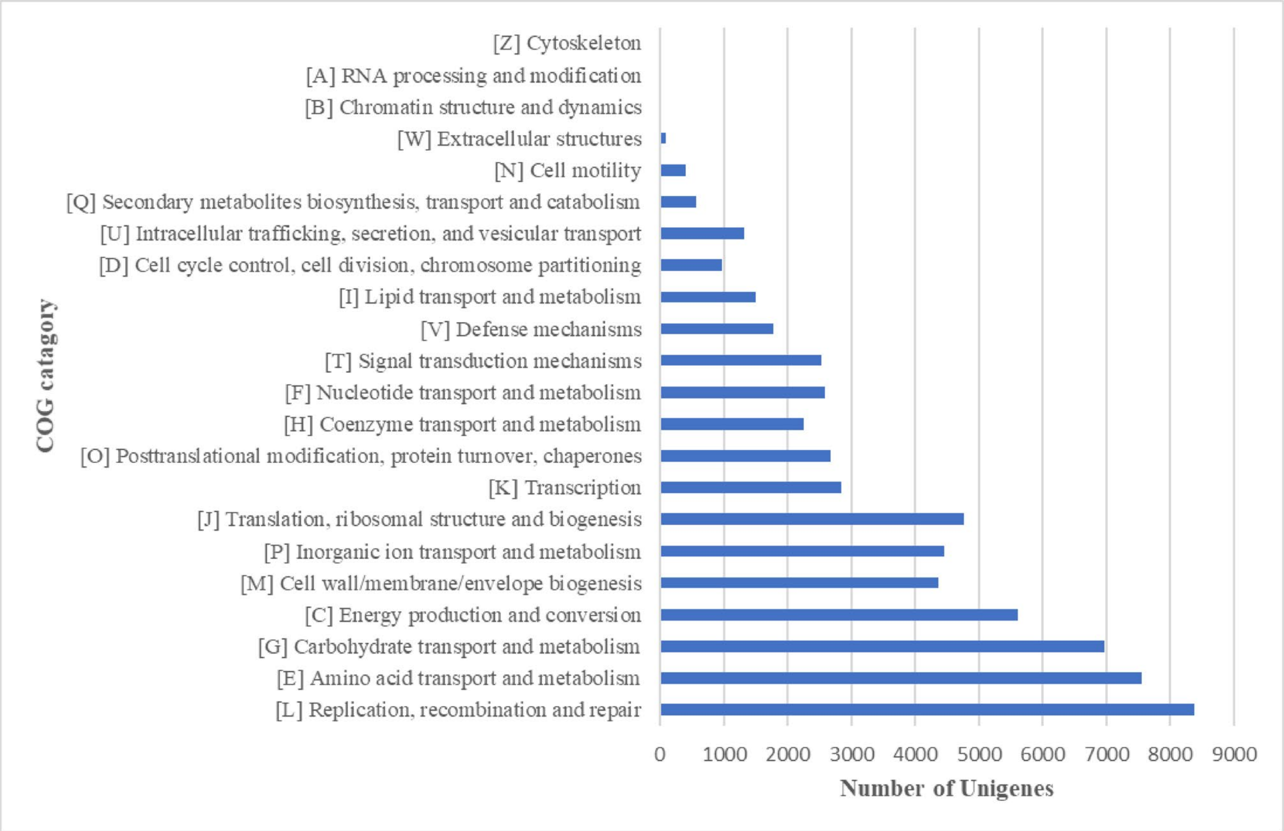

Fig. 7. Clustered bar chart representing the clusters of orthologous groups of proteins functional classification by using evolutionary genealogy of genes: non-supervised orthologous groups classification technique.

Most of the identified AMR genes belong to beta-lactamase. As shown in Fig. 9, the major bacterial phyla carrying AMR genes include Firmicutes (36%), Proteobacteria (31%), Actinobacteria (21%), and Bacteroidetes (5%). Pathogenic and opportunistic bacteria, such as *Escherichia coli*, *Mycobacterium tuberculosis*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Enterobacter cloacae*, *Enterococcus faecalis*, *Enterococcus faecium*, *Salmonella enterica*, *Paenibacillus* sp, *Neisseria gonorrhoeae*, *Acinetobacter baumannii*, *Enterococcus gallinarum*, *Rhodococcus equi*, *Streptococcus pyogenes*, etc. were found to the host multiplying AMR genes. Further, the AMR genes are grouped as peptide (10%), penam (9%), cephalosporin (9%), carbapenem (7%), aminoglycoside (6%), glycopeptide (6%), rifamycin (4%), fluoroquinolone (4%), macrolide (4%), cephamycin (4%), phenicol (3%), tetracycline (3%), Fosfomycin (2%), triclosan (2%), monobactam (2%), lincosamide (2%), aminocoumarin (2%), streptogramin (2%), penem (2%), isoniazid (2%), polyamine (2%), fusidic acid (2%), pleuromutilin (2%), para-aminosalicylic acid (2%), etc. Some AMR genes identified mobile genetic elements like integrons, transposons, colistin, and plasmids, suggesting the high potential for AMR properties transfer within the farm environment[39]. The transfer of AMR genes within the farm environment is possible through a horizontal gene transfer mechanism due to the contamination of DWW in soil[40]. DWW serves as a significant reservoir for AMR genes, which can lead to resistome enrichment and soil contamination when carried out on land. The AMR genes and mobile genetic elements are correlated, resulting in horizontal gene transfer among different microorganisms[41]. The destruction of the mobile genetic elements can inhibit the emergence of the waterborne AMR genes. The steps of WW treatment techniques must comply with such destruction[42].

VF abundance

The VFs and their corresponding classes were accessed to evaluate the prevalence of VGs in the location of the dairy industry, which is aware of the possible pathogenic risks in its surrounding environment. These genes indicated the distribution of potentially harmful microorganisms in the immediate surroundings. There was a substantial diversity of VGs at the location, as 2795 genes belonging to 11 virulence categories were identified from the DWW sample (Table 8). The 'immune modulation' virulence category was the most frequently detected, with 762 genes, followed by 'nutritional/metabolic factor' (490 genes), 'motility' (329 genes), 'adherence' (314 genes), 'effector delivery system' (366 genes), 'regulation' (211 genes), 'biofilm' (103 genes), 'stress survival' (77 genes), 'exoenzyme' (56 genes), 'invasion' (55 genes), 'antimicrobial activity/competitive advantage' (32 genes). The DWW sample accommodated a large diversity of different VGs, including papR, pgaC, and mucD, that are involved with biofilm formation: msbA, SAK_RS06335, gale, cpsA/uppS, rmlB, glf, gndA, BJAB07104_RS00505, acpXL, rfbB, cpsI, and manA that are involved with immune modulation; mbtI, hemB, hemL, hemC, carB, sugC, mgtB, BJE04_RS21750, ctpV, and hitC that are involved with nutritional/metabolic factor; groEL, ppk1,

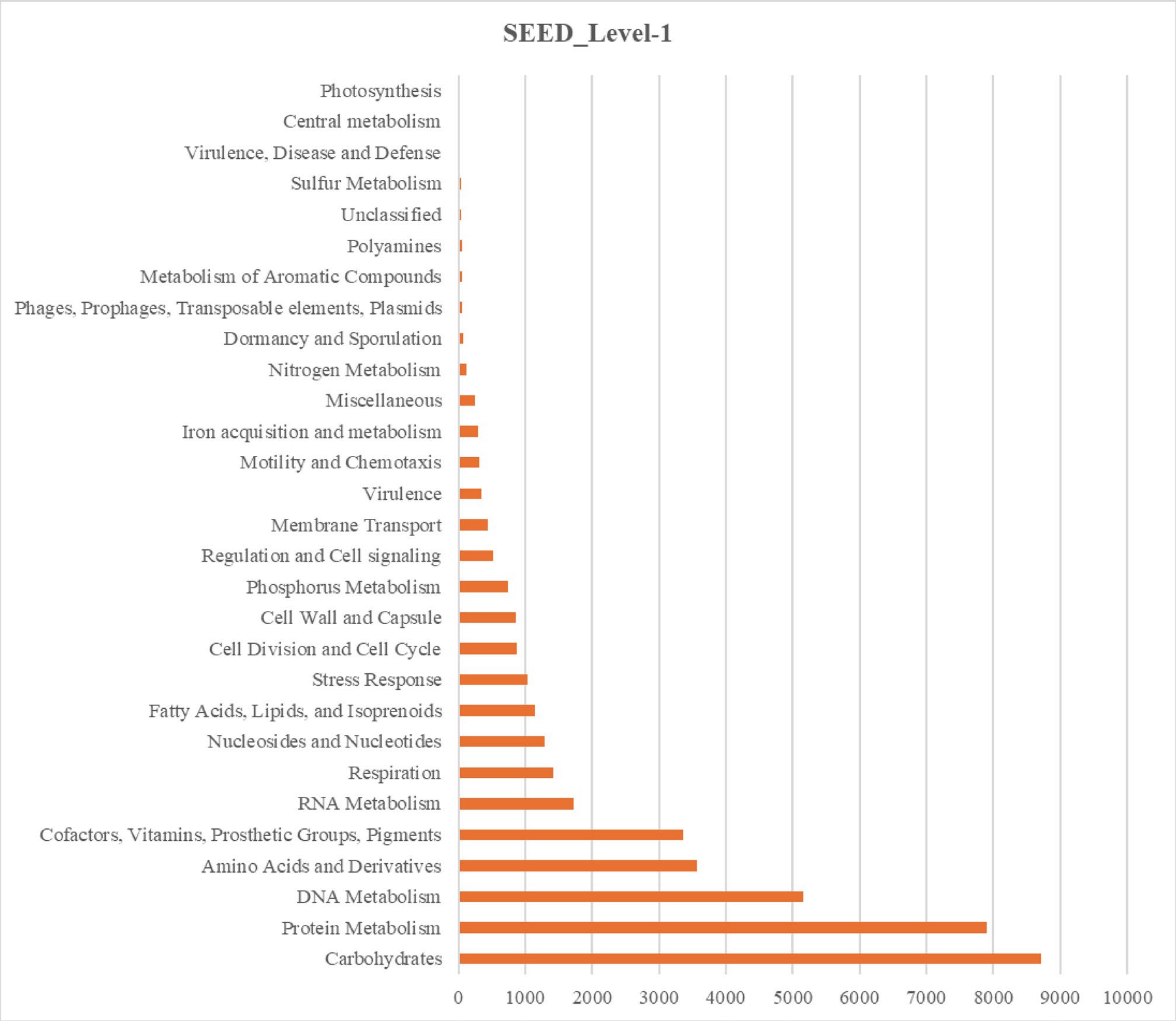

Fig. 8. Level-1 SEED classification for the functional behavior of metagenomic data sets obtained using DIAMOND blast tool.

IlpA, tufa, plr/gapA, htpB, pilR, lepB, CPE_RS09090, srtC-2/srtC, and lap that are involved with adherence; clpV, lirB, iagB, and ricA that are involved with effector delivery system; sigA/rpoV and gacS that are involved with regulation; eno, htrA/degP, srtA, and cppA that are involved with exoenzyme; iap/cwhA, kpsU, aut, and ibeB that are involved with invasion; ureB, sodB, katA, and tig/ropA that are involved with stress survival; acrB, mtrD, farB, and farA that are involved with antimicrobial activity/competitive advantage. The study showed that genes linked to flagella, like flmH, cheD, cheV3, flrA, tar, flea, TSR, cheB, fliC, flgR, etc., are widely distributed in the DWW sample. Additionally, a small number of VGs were seen, primarily associated with immunological regulation (acpXL) and twitching motility (pilG), indicating that environmental stresses strongly impact the distribution of VF.

The chemical and biological properties of WW allow its microbial communities to attach to the host tissues and abiotic components, initiating colonization followed by biofilm formation[43]. Different motility structures, aided by the flagella of the microorganisms, help in suitable environments for biofilm formation. Biofilms prevent the immune response from the host and environment, resulting in the development of different chronic diseases[44,45]. The patterns of gene expression change and evolve as infection progresses[46]. The chronic infections developed by biofilm are further increased by immune modulation. Bacterial pathogens use invasion mechanisms to enter host cells and tissues. Evasion from the immune system is made possible by intracellular survival. Bacteria can multiply inside host cells and evade immune detection through invasion. Pathogens produce extracellular enzymes such as lipases, nucleases, and proteases, aiding bacterial invasion, nutrition uptake, and immune response evasion[47]. Motility promotes biofilm development and invasion. Pathogens use motility to move away from hazardous environments and toward food security, determining the VFs[48]. Many bacteria manipulate host cell functions to their advantage by directly injecting effector proteins into the cells through secretion systems.

Resistance	AMR genes family	Organisms
Aminocoumarin; Carbapenem; Peptide; Rifamycin	ATP-binding cassette (ABC); Efflux pump	*Klebsiella pneumonia*; *Escherichia coli*
Diaminopyrimidine; Fluoroquinolone; Phenicol	Resistance-nodulation-cell division (RND); Efflux pump	*Escherichia coli*; *Pseudomonas aeruginosa*; *Klebsiella pneumoniae*; *Enterobacter cloacae*
Aminoglycoside; Cephalosporin; Macrolide; Peptide; Rifamycin; Tetracycline	Major facilitator superfamily (MFS), Efflux pump	*Klebsiella pneumoniae*; *Escherichia coli*
Elfamycin	Elfamycin resistant EF-Tu	*Enterococcus faecium*; *Streptomyces cinnamoneus*; *Escherichia coli*
Fluoroquinolone	Fluoroquinolone resistant gyrA	*Burkholderia dolosa*; *Neisseria gonorrhoeae*
Fusidic acid	Resistant fusA	*Staphylococcus aureus*
Peptide	pmr phosphoethanolamine transferase	*Escherichia coli*; *Klebsiella pneumoniae*
Peptide; Rifamycin	Rifamycin-resistant beta-subunit of RNA polymerase (rpoB)	*Escherichia coli*
Aminocoumarin; Fluoroquinolone	Fluoroquinolone resistant gyrB	*Clostridium ljungdahlii*
Fosfomycin	Resistant murA transferase	*Escherichia coli*
Mupirocin	Resistant isoleucyl-tRNA synthetase (ileS)	*Bifidobacterium bifidum*
Glycopeptide	Vancomycin-resistant beta prime subunit of RNA polymerase (rpoC)	*Clostridioides difficile*
Pyrazinamide	Pyrazinamide resistant rpsA	*Mycobacterium tuberculosis*
Fusidic acid	Resistant fusE	*Staphylococcus aureus*
Triclosan	Resistant fabG	*Escherichia coli* str. K-12 substr. MG1655
Carbapenem; Cephalosporin; Cephamycin; Monobactam; Penam	Penicillin-binding protein mutations conferring resistance to beta-lactam	*Streptococcus pneumoniae*
Tetracycline	Tetracycline-resistant ribosomal protection protein	*Bacteroides fragilis*
Peptide	Daptomycin resistant CdsA	*Streptococcus oralis*
Peptide; Rifamycin	Daptomycin resistant beta-subunit of RNA polymerase (rpoB)	*Staphylococcus aureus*
Peptide	Daptomycin resistant liaR	*Enterococcus faecium*
Glycopeptide	Glycopeptide resistance gene cluster; vanU	*Enterococcus faecalis*
Glycopeptide	Glycopeptide resistance gene cluster; vanR	*Rhodococcus equi*
Gosfomycin	Resistant ptsI phosphotransferase	*Escherichia coli*
Lincosamide; Macrolide; Oxazolidinone; Phenicol; Pleuromutilin; Streptogramin; Tetracycline	ABC-F ATP-binding cassette ribosomal protection protein	*Streptococcus pyogenes*
Peptide	Daptomycin-resistant beta prime subunit of RNA polymerase (rpoC)	*Staphylococcus aureus*
Para-aminosalicylic acid	Aminosalicylate resistant thymidylate synthase	*Mycobacterium tuberculosis*
Aminocoumarin	Aminocoumarin resistant gyrB	*Escherichia coli*
Aminoglycoside	kdpDE	*Escherichia coli*
Aminoglycoside	Resistant tlyA	*Mycobacterium tuberculosis*
Fluoroquinolone	Fluoroquinolone resistant parC	*Salmonella enterica*; *Acinetobacter baumannii*
Sulfonamide; Sulfone	Sulfonamide resistant dihydropteroate synthase folP	*Streptococcus pyogenes*; *Neisseria gonorrhoeae*; *Escherichia coli*
Streptogramin	Streptogramin vat acetyltransferase	*Yersinia enterocolitica*
Aminoglycoside	Resistant rpsL	*Mycobacterium tuberculosis*
Peptide	Undecaprenyl pyrophosphate related proteins	*Escherichia coli*
Pleuromutilin	Ribosomal protein mutation conferring resistance to pleuromutilin	*Thermus thermophilus*
Peptide	Daptomycin resistant pgsA	*Staphylococcus aureus*
Carbapenem; Cephalosporin; Cephamycin; Fluoroquinolone; Glycylcycline; Monobactam; Penam; Penem; Phenicol; Rifamycin; Tetracycline; Triclosan	General Bacterial Porin with reduced permeability to beta-lactams	*Enterobacter cloacae*
Tetracycline	Tetracycline inactivation enzyme	Uncultured bacterium
Lincosamide	Llm 23 S ribosomal RNA methyltransferase	*Paenibacillus* sp. LC231
Prothionamide	Prothionamide resistant katG	*Mycobacterium tuberculosis*
Fluoroquinolone	Fluoroquinolone resistant parE	*Escherichia coli*
Isoniazid; triclosan	Resistant fabI	*Escherichia coli*
Peptide	Daptomycin resistant gshF	*Enterococcus faecalis*
Peptide	Daptomycin resistant walK	*Staphylococcus aureus*
Peptide	Acinetobacter mutant Lpx gene conferring resistance to colistin	*Acinetobacter baumannii*
Aminocoumarin	Aminocoumarin resistant parE	*Staphylococcus aureus*
Triclosan	Triclosan resistant gyrA	*Salmonella enterica*
Fosfomycin	Resistant UhpT	*Escherichia coli*
Continued		

Resistance	AMR genes family	Organisms
Glycopeptide	Glycopeptide resistance gene cluster; vanH	*Enterococcus faecium*
Fluoroquinolone	Multidrug and toxic compound extrusion (MATE) transporter	*Salmonella enterica*
Phenicol	Chloramphenicol acetyltransferase (CAT)	*Enterobacter cloacae*; *Agrobacterium tumefaciens*
Glycopeptide	Van ligase	*Enterococcus gallinarum*
Fosfomycin	Resistant GlpT	*Escherichia coli*
Macrolide	Small multidrug resistance (SMR); efflux pump	*Escherichia coli*
Cephalosporin; Penam	ampC-type beta-lactamase	*Escherichia coli*
Glycopeptide	murG transferase	*Clostridium acetobutylicum*
Zoliflodacin; Aminocoumarin	Zoliflodacin resistant gyrB	*Neisseria gonorrhoeae*
Isoniazid; Triclosan	Resistant kasA	*Mycobacterium tuberculosis*
Peptide	Daptomycin resistant YybT	*Enterococcus faecalis*
Para-aminosalicylic acid	Aminosalicylate resistant dihydrofolate synthase	*Mycobacterium tuberculosis*
Rifamycin	Rifampin Phosphotransferase	*Paenibacillus* sp. LC231
Sulfonamide; Sulfone	Dapsone resistant dihydropteroate synthase folp	*Mycobacterium leprae*
Glycopeptide	Glycopeptide resistance gene cluster; vant	*Enterococcus gallinarum*
Sulfonamide	Sulfonamide resistant sul	uncultured bacterium
Fosfomycin	Resistant cya adenylate cyclase	*Escherichia coli*
Glycopeptide	Glycopeptide resistance gene cluster; vany	*Enterococcus faecium*
Fosfomycin	Fosfomycin thiol transferase	*Escherichia coli*
Aminoglycoside	Resistant gidb	*Mycobacterium tuberculosis* H37Rv complete genome
Peptide	Daptomycin resistant cls	*Enterococcus faecalis*
Fluoroquinolone; tetracycline	Sugar porin (sp)	*Escherichia coli* O157:H7 str. Sakai DNA; complete genome
Diaminopyrimidine	Trimethoprim resistant dihydrofolate reductase dfr	*Escherichia coli strain* F123 class 1 integron; partial sequence
Carbapenem; Cephalosporin; Cephamycin; Penam	Nmca beta-lactamase	*Enterobacter cloacae*
Acridine dye; Aminoglycoside; Carbapenem; Cephalosporin; Cephamycin; Disinfecting Agents and intercalating dyes; Fluoroquinolone; Macrolide; Monobactam; Penam; Penem; Phenicol; Tetracycline	Outer membrane porin (opr); Resistance-nodulation-cell division (rnd); Efflux pump	*Pseudomonas aeruginosa*
Glycopeptide; Macrolide	50 S rRNA with mutation conferring resistance to macrolide	*Neisseria gonorrhoeae*
Nitrofuran	Resistant nfsA	*Escherichia coli*
Aminoglycoside	ANT(6)	*Exiguobacterium* sp. S3-2
Glycopeptide	Glycopeptide resistance gene cluster; vanZ	*Paenibacillus popilliae*
Aminoglycoside; Fluoroquinolone	MipA-interacting protein	*Escherichia coli*
Rifamycin	Rifampin monooxygenase	*Rhodococcus equi*
Pyrazinamide	Pyrazinamide resistant pncA	*Mycobacterium tuberculosis*
Aminoglycoside	Phenotypic variant regulator	*Pseudomonas aeruginosa*
Peptide	MCR phosphoethanolamine transferase	*Klebsiella pneumoniae*, *Salmonella enterica*
Glycopeptide	Glycopeptide resistance gene cluster; vanW	*Desulfitobacterium hafniense*
Lincosamide; Macrolide	Non-erm 23 S ribosomal RNA methyltransferase (G748)	*Micromonospora griseorubida*
Prothionamide	Prothionamide resistant mshA	*Mycobacterium tuberculosis*
Aminoglycoside	AAC(6')	*Enterobacter cloacae*
isoniazid	resistant ndh	*Mycobacterium smegmatis*
Glycopeptide	Glycopeptide resistance gene cluster; vanS	*Enterococcus gallinarum*
Lincosamide; Macrolide; Streptogramin	Erm 23s ribosomal rna methyltransferase	*Prescottella equi*
Glycopeptide	Glycopeptide resistance gene cluster; vanXY	*Enterococcus faecalis*
Peptide	Intrinsic peptide; Resistant Lps	*Acinetobacter baumannii*
Penam	CBP beta-lactamase	*Clostridium botulinum*
Fluoroquinolone	Quinolone resistance protein (qnr)	*Salmonella enterica*
Rifamycin	Rifampin ADP-ribosyltransferase (Arr)	*Pseudomonas aeruginosa*
Peptide	Bah amidohydrolase	*Paenibacillus* sp. LC231
Ethionamide; Isoniazid; Prothionamide	Prothionamide resistant ethA	*Mycobacterium tuberculosis*
Cephalosporin; monobactam	MIR beta-lactamase	*Enterobacter asburiae*
Carbapenem; Cephalosporin; Penam	SHV beta-lactamase	*Escherichia coli*
Carbapenem; Cephalosporin; Penam	OXA beta-lactamase	*Pseudomonas aeruginosa*
Continued		

Resistance	AMR genes family	Organisms
Glycopeptide	Glycopeptide resistance gene cluster; vanX	*Desulfitobacterium hafniense*
Cephalosporin; penam	OKP beta-lactamase	*Klebsiella pneumoniae*
Penam; Penem	LEN beta-lactamase	*Klebsiella pneumoniae*
Rifamycin	RbpA bacterial RNA polymerase-binding protein	*Mycobacterium smegmatis*
Peptide	Daptomycin resistant liaS	*Enterococcus faecalis*
Carbapenem; Cephalosporin; Cephamycin; Monobactam; Penam	Methicillin resistant PBP2	*Staphylococcus aureus*
Aminoglycoside	APH(3')	Plasmid RP4
Phenicol	Chloramphenicol phosphotransferase	*Streptomyces venezuelae*
Lincosamide; Oxazolidinone; Phenicol; Pleuromutilin; Streptogramin	Cfr 23 S ribosomal RNA methyltransferase	*Paenibacillus* sp. Y412MC10
Peptide	Defensin resistant mprF	*Listeria monocytogenes*
Nucleoside	Tunicamycin resistance protein	*Bacillus subtilis*
Glycopeptide	Glycopeptide resistance gene cluster; vanV	*Enterococcus faecalis*
Carbapenem; Cephalosporin; Cephamycin; Penam	ACT beta-lactamase	*Enterobacter asburiae*
Nucleoside	Streptothricin acetyltransferase (SAT)	*Escherichia coli*
Aminoglycoside	ANT(9)	*Staphylococcus aureus*
Polyamine	Ethambutol resistant embC	*Mycobacterium tuberculosis*
Penam	AER beta-lactamase	*Aeromonas hydrophila*
Carbapenem	CarO porin	*Acinetobacter baumannii*
Lincosamide	Lincosamide nucleotidyltransferase (LNU)	*Streptococcus uberis*
Cephalosporin; Penam; Penem	ACI beta-lactamase	*Acidaminococcus fermentans*
Macrolide	mgt macrolide glycotransferase	*Streptomyces lividans*
Carbapenem; cephalosporin	CIA beta-lactamase	*Chryseobacterium indologenes*
Peptide	Lipid A phosphatase	*Porphyromonas gingivalis*
Cephalosporin	CTX-M beta-lactamase	*Klebsiella pneumoniae*
Carbapenem	BJP beta-lactamase	*Bradyrhizobium diazoefficiens*
Carbapenem	IND beta-lactamase	*Chryseobacterium indologenes*
Cephalosporin	CMH beta-lactamase	*Enterobacter cloacae*
Peptide	daptomycin resistant liaF	*Enterococcus faecium*
Aminoglycoside	APH(6)	*Streptomyces glaucescens*
Isoniazid; Triclosan	isoniazid resistant inhA	*Mycobacterium tuberculosis*
Cephalosporin; Cephamycin; Penam	AIM beta-lactamase	*Pseudomonas aeruginosa*
Streptogramin	Streptogramin vgb lyase	*Paenibacillus* sp. LC231
Polyamine	Ethambutol resistant embA	*Mycobacterium tuberculosis*
Peptide	Lysocin resistant menA	*Staphylococcus aureus*
Cephalosporin	BAT Beta-lactamase	*Bacillus atrophaeus*
Macrolide	Ole glycosyltransferase	*Streptomyces*
Peptide	Intrinsic colistin resistant phosphoethanolamine transferase	*Moraxella osloensis*
Cephalosporin; cephamycin	DHA beta-lactamase	*Enterobacter cloacae*
Aminoglycoside	ANT(3")	*Escherichia coli*
Carbapenem; penam	CAU beta-lactamase	*Caulobacter crescentus*
Aminoglycoside	AAC(3)	*Streptomyces paromomycinus*
Aminoglycoside	ANT(4')	*Pseudomonas aeruginosa*
Cephalosporin; cephamycin; monobactam; penam; penem	OCH beta-lactamase	*Ochrobactrum anthropi*
Penam	BCL Beta-lactamase	*Bacillus clausii*
Cephalosporin; Penam	Class A LRA beta-lactamase	Uncultured bacterium
Carbapenem	GPC beta-lactamase	*Pseudomonas aeruginosa*
Carbapenem	CphA beta-lactamase	*Aeromonas veronii*
Rifamycin	Rifampin glycosyltransferase	*Streptomyces* sp. WAC1438
Para-aminosalicylic acid	Aminosalicylate resistant dihydrofolate reductase	*Mycobacterium tuberculosis*
Fusidic acid	Target protecting FusB-type protein conferring resistance to Fusidic acid	*Staphylococcus saprophyticus*
Carbapenem	BKC Beta-lactamase	*Klebsiella pneumoniae*
Continued		

Resistance	AMR genes family	Organisms
Cephalosporin	AQU beta-lactamase	*Aeromonas dhakensis*
Polyamine; Rifamycin	Rifamycin-resistant arabinosyltransferase	*Mycobacterium tuberculosis*
Macrolide	Macrolide phosphotransferase (MPH)	*Escherichia coli*
Carbapenem	SPG beta-lactamase	*Sphingomonas* sp. ALS-13
Polyamine	Ethambutol resistant iniA	*Mycobacterium tuberculosis*

Table 7. Results of the AMR genes and linked organisms.

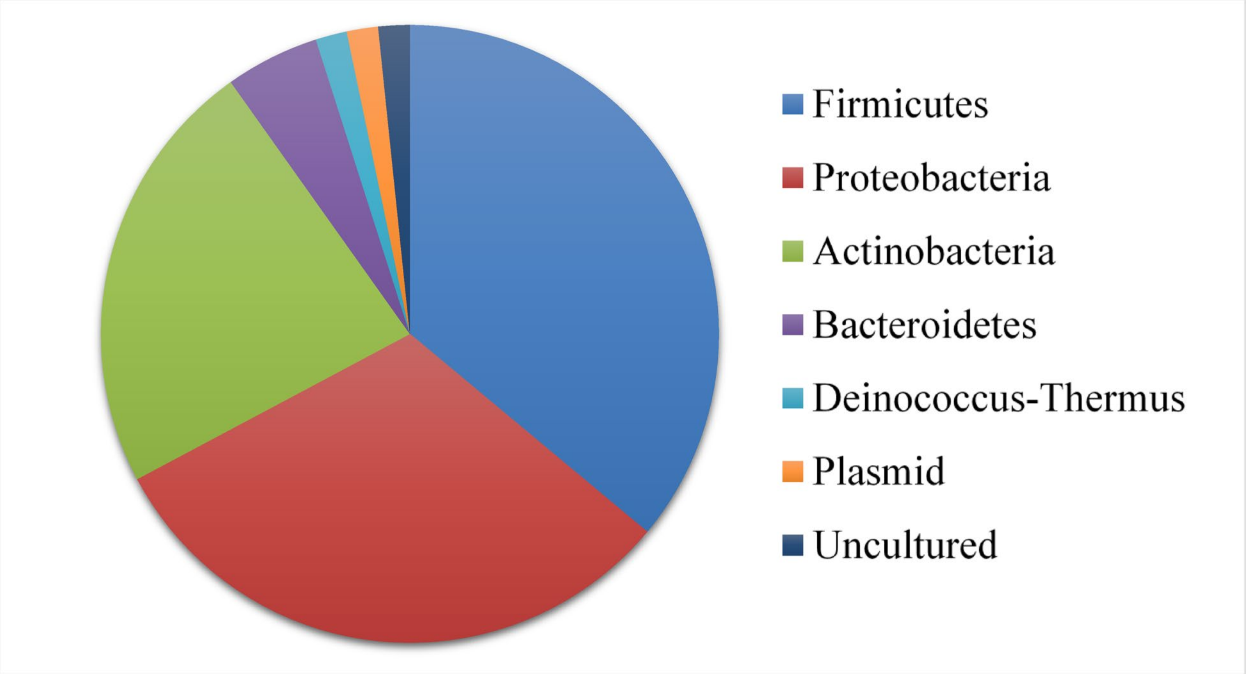

Fig. 9. Major bacterial phyla carrying AMR genes.

Effectors alter the cytoskeleton, signalling cascades, and immunological responses of host cells. Many Gram-negative bacteria depend on them for their pathogenicity[49]. Quorum sensing and environmental cues are two-component systems that closely control the expression of VFs, ensuring that bacteria only express them when necessary. One strategy for creating anti-virulence treatments is to target regulatory mechanisms. Pathogens cannot withstand host-induced stressors like oxidative stress, low pH, limited nutrients, reactive oxygen species, and reactive nitrogen species[47].

Conclusion

The DWW sample under investigation in this article contained high organic and inorganic nutrients, as its BOD, COD, TSS, and TP content were above the environmental permissible limits. This result showed an increased amount of proteins, fats, and carbohydrates in the form of coagulated milk, cheese curd fines, and flavouring ingredients. The BOD/COD ratio revealed a higher pollutant level in the DWW sample and its poor biodegradability. The DNA concentration above 100 ng/µl confirmed that the DWW contained large microbial diversity with 2,211,670 raw reads. The DWW harboured a complex microbial community with microorganisms belonging to 36 phyla, 72 classes, 111 orders, 168 families, 275 genera, and 347 species. The DWW sample contained 94.5% bacterial communities, followed by Eukaryota (3.2%), archaea (1.7%), and viruses (0.3%). Firmicutes were the most dominant bacteria at the phylum level, with 51% abundance. At the class level, Negativicutes were most predominant with 25%. At the family level, Selenomonadaceaes were most dominant, with 22%. Archaea like *Methanoregula boonei*, *Methanospirillum*, *Methanosaeta*, and *Methanomassiliicoccales* were found in the DWW sample. The Eukaryota community included many important fungal and microalgae species in the DWW. Some unclassified viral species like *Siphoviridae*, *Myoviridae*, *Podoviridae*, *Caudovirales*, and unidentified phages were identified in the DWW sample. Shannon index and Simpson index showed that the DWW sample has a high level of microbial diversity in a few species. In the GO category level-2, a total of 481 hits were found for molecular function, cellular component, and biological process categories. In the GO category level-3, a total of 3151 hits were found for molecular function, cellular component, and biological process categories. The COG analysis revealed the functional roles of a wide range of genes; however, genes

NCBI Accession No	Virulence gene	Virulence factor	Virulance class
WP_000685095	SPD_RS01770	Nucleotide sugar dehydrogenase [Capsule (VF0144)] [*Streptococcus pneumoniae* D39]	Immune modulation (VFC0258)
WP_005687513	msbA	Lipid transporter ATP-binding/permease [LOS (VF0044)] [*Haemophilus influenzae* PittEE]	Immune modulation (VFC0258)
WP_001222601	SAK_RS06335	LysR family transcriptional regulator [Capsule (VF0274)] [*Streptococcus agalactiae* A909]	Immune modulation (VFC0258)
WP_000996524	galE	UDP-glucose 4-epimerase GalE [Polysaccharide capsule (VF0659)] [*Bacillus cereus* AH187]	Immune modulation (VFC0258)
WP_002294134	cpsA/uppS	Undecaprenyl diphosphate synthase [Capsule (VF0361)] [*Enterococcus faecium* Aus0004]	Immune modulation (VFC0258)
WP_011261022	rmlB	dTDP-glucose 4,6-dehydratase [Capsular polysaccharide (VF0624)] [*Vibrio fischeri* ES114]	Immune modulation (VFC258)
YP_002344822	glf	UDP-galactopyranose mutase [Capsule (VF0323)] [*Campylobacter jejuni* subsp. *jejuni* NCTC 11168]	Immune modulation (VFC258)
WP_015958700	gndA	NADP-dependent phosphogluconate dehydrogenase [Capsule (VF0560)] [*Klebsiella pneumoniae* subsp. *pneumoniae* MGH 78578]	Immune modulation (VFC258)
WP_000705810	BJAB07104_RS00505	UDP-glucose/GDP-mannose dehydrogenase family protein [Capsule (VF0465)] [*Acinetobacter baumannii* BJAB07104]	Immune modulation (VFC0258)
WP_002963616	acpXL	Acyl carrier protein [LPS (VF0367)] [*Brucella melitensis* bv. 1 str. 16 M]	Immune modulation (VFC0258)
WP_002947383	rfbB	dTDP-glucose 4,6-dehydratase [Capsule (VF0144)] [*Streptococcus thermophilus* LMD-9]	Immune modulation (VFC0258)
WP_002376666	cpsI	UDP-galactopyranose mutase [Capsule (VF0361)] [*Enterococcus faecalis* V583]	Immune modulation (VFC0258)
WP_000276321	manA	Mannose-6-phosphate isomerase, class I [Polysaccharide capsule (VF0659)] [*Bacillus thuringiensis* str. Al Hakam]	Immune modulation (VFC0258)
YP_007957699	mbtI	Isochorismate synthase MbtI [Mycobactin (VF0299)] [*Mycobacterium tuberculosis* str. Haarlem/NITR202]	Nutritional/Metabolic factor (VFC0272)
WP_011608705	hemB	Porphobilinogen synthase [Heme biosynthesis (VF0758)] [*Haemophilus somnus* 2336]	Nutritional/Metabolic factor (VFC0272)
WP_011609384	hemL	Glutamate-1-semialdehyde 2,1-aminomutase [Heme biosynthesis (VF0758)] [*Haemophilus somnus* 129PT]	Nutritional/Metabolic factor (VFC0272)
WP_011608203	hemC	Hydroxymethylbilane synthase [Heme biosynthesis (VF0758)] [*Haemophilus somnus* 2336]	Nutritional/Metabolic factor (VFC0272)
WP_014547360	carB	Carbamoyl phosphate synthase large subunit [Pyrimidine biosynthesis (VF0558)] [*Francisella cf. tularensis* subsp. *novicida* 3523]	Nutritional/Metabolic factor (VFC0272)
WP_005098896	sugC	Sn-glycerol-3-phosphate ABC transporter ATP-binding protein UgpC [Trehalose-recycling ABC transporter (VF0842)] [*Mycobacterium abscessus* subsp. *bolletii* 50594]	Nutritional/Metabolic factor (VFC0272)
WP_000131305	mgtB	Mg2 + transport protein [MgtBC (VF0106)] [*Salmonella enterica* subsp. *arizonae serovar* 62:z4,z23:-- str. RSK2980]	Nutritional/Metabolic factor (VFC0272)
WP_011152542	BJE04_RS21750	3-Deoxy-7-phosphoheptulonate synthase [Vibriobactin/Vulnibactin (VF0626)] [*Vibrio vulnificus* YJ016]	Nutritional/Metabolic factor (VFC0272)
WP_008257346	ctpV	Copper-translocating P-type ATPase [Copper exporter (VF0849)] [*Mycobacterium indicus pranii* MTCC 9506]	Nutritional/Metabolic factor (VFC0272)
WP_011608934	hitC	Iron(III) ABC transporter, ATP-binding protein [HitABC (VF0268)] [*Haemophilus somnus* 129PT]	Nutritional/Metabolic factor (VFC0272)
WP_005300916	flmH	Short chain dehydrogenase/reductase family oxidoreductase [Polar flagella (VF0473)] [*Aeromonas hydrophila* ML09-119]	Motility (VFC0204)
WP_005168769	cheD	Methyl-accepting chemotaxis protein CheD [Flagella (VF0394)] [*Yersinia enterocolitica* subsp. *enterocolitica* 8081]	Motility (VFC0204)
WP_000467802	cheV3	Chemotaxis coupling protein CheV3 [Flagella (VF0051)] [*Helicobacter pylori* 26695]	Motility (VFC0204)
WP_011262363	flrA	Sigma-54 dependent transcriptional activator [Flagella (VF0519)] [*Vibrio fischeri* ES114]	Motility (VFC0204)
WP_010904770	tar	Methyl-accepting chemotaxis protein II [Peritrichous flagella (VF1154)] [*Escherichia coli* O157:H7 str. EDL933]	Motility (VFC0204)
WP_010946589	fleQ	Transcriptional regulator FleQ [Flagella (VF0157)] [*Legionella pneumophila* subsp. *pneumophila* str. Philadelphia 1]	Motility (VFC0204)
WP_003514589	groEL	Chaperonin GroEL [GroEL (VF0594)] [*Clostridium thermocellum* ATCC 27405]	Adherence (VFC0001)
WP_012031410	ppk1	Polyphosphate kinase 1 [Type IV pili (VF1212)] [*Dichelobacter nodosus* VCS1703A]	Adherence (VFC0001)
WP_011149697	IlpA	Immunogenic lipoprotein A [IlpA (VF0513)] [*Vibrio vulnificus* YJ016]	Adherence (VFC0001)
WP_014714676	tufA	Elongation factor Tu [EF-Tu (VF0460)] [*Francisella noatunensis* subsp. *orientalis* str. Toba 04]	Adherence (VFC0001)
WP_000260656	plr/gapA	Type I glyceraldehyde-3-phosphate dehydrogenase [Streptococcal plasmin receptor/GAPDH (VF1042)] [*Streptococcus agalactiae* A909]	Adherence (VFC0001)
WP_197532029	htpB	Hsp60, 60 K heat shock protein HtpB [Hsp60 (VF0159)] [*Legionella longbeachae* NSW150]	Adherence (VFC0001)
WP_012722471	pilR	Two-component response regulator PilR [Type IV pili (VF0082)] [*Pseudomonas fluorescens* SBW25]	Adherence (VFC0001)
WP_000828633	lepB	Signal peptidase I [RlrA islet (VF0529)] [*Streptococcus agalactiae* COH1]	Adherence (VFC0001)
Continued			

NCBI Accession No	Virulence gene	Virulence factor	Virulance class
WP_003451114	CPE_RS09090	Type IV pilus twitching motility protein PilT [Type IV pili (VF0731)] [*Clostridium perfringens* str. 13]	Adherence (VFC0001)
WP_011837256	srtC-2/srtC	sortase [RlrA islet (VF0529)] [*Streptococcus sanguinis* SK36]	Adherence (VFC0001)
WP_014092990	lap	Listeria adhesion protein Lap [Lap (VF0444)] [*Listeria ivanovii* subsp. *ivanovii* PAM 55]	Adherence (VFC0001)
WP_011081076	VV1_RS15610	CpaF family protein [Flp pili (VF0612)] [*Vibrio vulnificus* CMCP6]	Adherence (VFC0001)
WP_011860976	CD0873	ABC transporter substrate-binding protein [CD0873 (VF0593)] [*Clostridium difficile* 630]	Adherence (VFC0001)
WP_004890358	clpV	ATP-dependent chaperone ClpB [T6SS-II (VF0783)] [*Klebsiella oxytoca* E718]	Effector delivery system (VFC0086)
WP_010947678	lirB	Dot/Icm type IV secretion system effector LirB [Dot/Icm T4SS secreted effectors (VF0798)] [*Legionella pneumophila* subsp. *pneumophila* str. Philadelphia 1]	Effector delivery system (VFC0086)
WP_011136188	iagB	Type III secretion system invasion protein IagB [Cpi-1a + Cpi-1 (SPI-1 like) (VF1259)] [*Chromobacterium violaceum* ATCC 12472]	Effector delivery system (VFC0086)
WP_002964382	ricA	Type IV secretion system effector RicA, Rab2 interacting conserved protein A [T4SS secreted effectors (VF0695)] [*Brucella melitensis* bv. 1 str. 16 M]	Effector delivery system (VFC0086)
YP_005923858	sigA/rpoV	RNA polymerase sigma factor [SigA (VF0257)] [*Mycobacterium tuberculosis* RGTB423]	Regulation (VFC0301)
WP_003156446	gacS	Two-component system sensor histidine kinase GacS [GacS/GacA two-component system (VF0908)] [*Pseudomonas aeruginosa* PA7]	Regulation (VFC0301)
WP_012130507	eno	Phosphopyruvate hydratase [Streptococcal enolase (VF1060)] [*Streptococcus gordonii* str. Challis substr. CH1]	Exoenzyme (VFC0251)
WP_000728356	htrA/degP	Trypsin-like peptidase domain-containing protein [Serine protease (VF1055)] [*Streptococcus agalactiae* A909]	Exoenzyme (VFC0251)
WP_011681303	srtA	Sortase [Sortase A (VF1048)] [*Streptococcus thermophilus* LMD-9]	Exoenzyme (VFC0251)
WP_000171847	cppA	CppA family protein [C3-degrading protease (VF1054)] [*Streptococcus agalactiae* NEM316]	Exoenzyme (VFC0251)
WP_010990501	iap/cwhA	P60 extracellular protein, invasion associated protein Iap [P60 (VF0068)] [*Listeria innocua* Clip11262]	Invasion (VFC0083)
WP_000030755	kpsU	3-Deoxy-manno-octulosonate cytidylyltransferase [K1 capsule (VF0239)] [*Escherichia coli* O45:K1:H7 str. S88]	Invasion (VFC0083)
WP_012985336	aut	Autolysin [Auto (VF0348)] [*Listeria seeligeri serovar* 1/2b str. SLCC3954]	Invasion (VFC0083)
WP_000074204	ibeB	Cu(+)/Ag(+) efflux RND transporter outer membrane channel CusC [Ibes (VF0237)] [*Escherichia coli* O157:H7 str. EDL933]	Invasion (VFC0083)
WP_011115250	ureB	Urease beta subunit UreB, urea amidohydrolase [Urease (VF0050)] [*Helicobacter hepaticus* ATCC 51449]	Stress survival (VFC0282)
WP_016357051	sodB	Superoxide dismutase [SodB (VF0169)] [*Legionella pneumophila* subsp. *pneumophila* str. Philadelphia 1]	Stress survival (VFC0282)
WP_003689973	katA	Catalase [KatA (VF0454)] [*Neisseria gonorrhoeae* NCCP11945]	Stress survival (VFC0282)
WP_000107753	tig/ropA	Trigger factor [Trigger factor (VF1056)] [*Streptococcus agalactiae* A909]	Stress survival (VFC0282)
WP_002264119	SMUNN2025_RS00190	CHAP domain-containing protein [Glucan-binding proteins (VF1046)] [*Streptococcus mutans* NN2025]	Biofilm (VFC0271)
WP_011199086	papR	DUF4084 domain-containing protein [PlcR-PapR quorum sensing (VF0660)] [*Bacillus cereus* E33L]	Biofilm (VFC0271)
WP_000866236	pgaC	Poly-beta-1,6 N-acetyl-D-glucosamine synthase [PNAG (VF0472)] [*Acinetobacter baumannii* BJAB0715]	Biofilm (VFC0271)
WP_011059790	mucD	Serine protease MucD precursor [Alginate (VF0091)] [*Pseudomonas fluorescens* Pf-5]	Biofilm (VFC0271)
WP_014228207	acrB	Acriflavine resistance protein B [AcrAB (VF0568)] [*Klebsiella oxytoca* E718]	Antimicrobial activity/Competitive advantage (VFC0325)
WP_013448560	mtrD	Multiple transferable resistance system protein MtrD [MtrCDE (VF0451)] [*Neisseria lactamica* 020 – 06]	Antimicrobial activity/Competitive advantage (VFC0325)
WP_002226030	farB	Fatty acid efflux system protein FarB [FarAB (VF0450)] [*Neisseria meningitidis* G2136]	Antimicrobial activity/Competitive advantage (VFC0325)
WP_013448426	farA	Fatty acid efflux system protein FarA [FarAB (VF0450)] [*Neisseria lactamica* 020 – 06]	Antimicrobial activity/Competitive advantage (VFC0325)

Table 8. Result of the virulence factor analysis.

involved in the cellular processes were found to be [L], [M], and [D]. A total of 155 resistance gene families, with 100 to 50% identical matches, were detected from the DWW sample; however, diverse categories of AMR genes were cephalosporin, carbapenem, and rifamycin. A total of 2795 genes belonging to 11 virulence categories were identified from the DWW sample. The study revealed that the DWW was highly polluted and contained several pathogenic microorganisms. The advantage of this DWW sample is that it contains several bacteria, fungi, and microalgae that the indigenous community can use to eliminate its pollutant load. However, complete screening of the microorganism for the particular DWW treatment may be the future course of this group to develop an in situ bioremediation process.

Data availability

The datasets generated during the current study are available in the NCBI BioProject database [Accession Number-PRJNA1238483].

Received: 12 March 2025; Accepted: 19 August 2025
Published online: 25 September 2025

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Acknowledgements
The authors are grateful to Prof. (Dr.) Manojranjan Nayak, President of Siksha 'O' Anusandhan (Deemed to be University) for providing infrastructure and encouragement.

Author contributions
Author contributions statement: o Swati Sucharita Satpathy: Experimental design; Conducting experiment; Data interpretation; Writing manuscripto Debabrata Pradhan: Data interpretation; Writing manuscript; Overall guidance.

Funding
Open access funding provided by Siksha 'O' Anusandhan (Deemed To Be University)

Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
Not applicable.

Consent for publication

The results/data/figures in this submission have not been previously published, or are not under consideration for publication elsewhere.

Additional information

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